SECTION 9 – MAGNET RAMPING

WARNING!

THE FOLLOWING REQUIRED SAFETY ACTIONS MUST BE TAKEN BEFORE RAMPING THE MAGNET:

1. **MAKE SURE MAGNET ROOM VENT EXHAUST FAN IS TURNED ON, OR THE HATCH IS OPENED IF A MOBILE VAN, BEFORE STARTING THIS PROCEDURE. THIS IS REQUIRED TO EXHAUST THE ODORLESS AND INVISIBLE HELIUM GAS GENERATED DURING THIS PROCEDURE AND PREVENT OXYGEN DISPLACEMENT IN THE MAGNET ROOM. REVIEW AND FOLLOW CRYOGEN SAFETY MEASURES CONTAINED IN SECTION 5–3 OF THE INTRODUCTION (CRYOGEN SAFETY).**

2. **NOTIFY SITE ADMINISTRATION BEFORE RAMPING THE MAGNET THAT ALL MAGNETIC SAFETY PRECAUTIONS MUST BE TAKEN.**

3. **POST WARNING SIGNS OUTSIDE THE 5 GAUSS ZONE TO ALERT PERSONNEL WITH CARDIAC PACEMAKERS, NEUROSTIMULATORS AND OTHER BIOSTIMULATION DEVICES NOT TO PROCEED INTO THE DESIGNATED AREA. POST THESE SIGNS ON THE MAGNET ROOM LEVEL AS WELL AS AREAS BELOW OR ABOVE THE MAGNET TO WHICH THE 5 GAUSS ZONE EXTENDS. SEE INTRODUCTION, SECTION 5.**

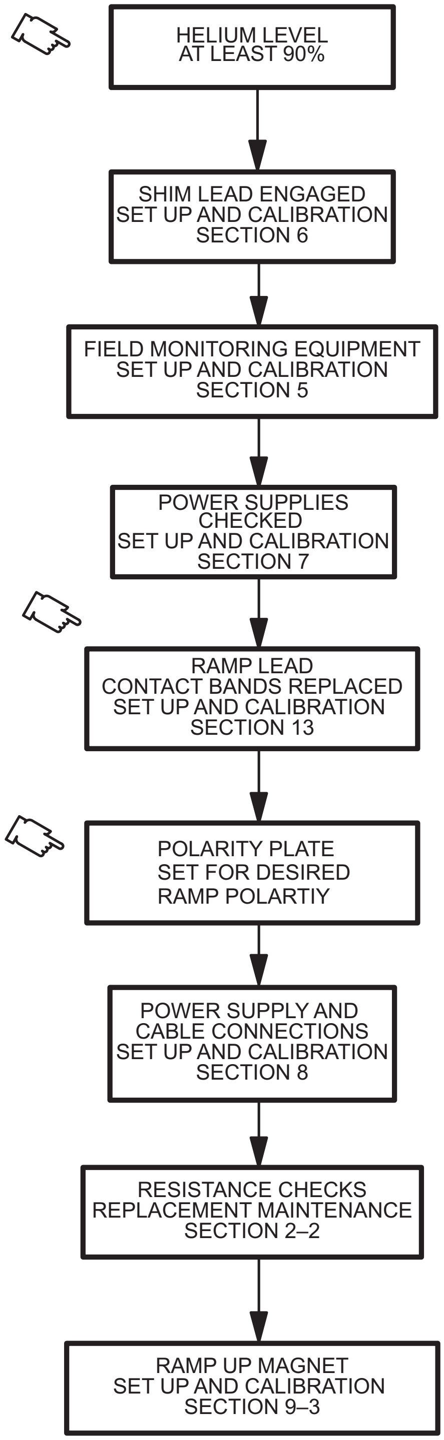
4. **REMOVE ALL LOOSE FERROMAGNETIC MATERIAL FROM THE MAGNET ROOM. PULL THE POWER SUPPLIES AS FAR AWAY FROM THE MAGNET AS THE CABLES AND SITE GEOMETRY ALLOW. METAL OBJECTS CAN BECOME DANGEROUS PROJECTILES IN A MAGNETIC FIELD.**

5. **MAKE SURE THAT THE MAGNET RUNDOWN UNIT IS INSTALLED AND OPERATIONAL TO ENABLE THE MAGNETIC FIELD TO BE QUICKLY DISCHARGED IN CASE OF AN EMERGENCY. SEE SET UP AND CALIBRATION, SECTION 1–5.**

6. **MAKE SURE THAT THE MAGNET IS AT LEAST 90% FULL OF LIQUID HELIUM TO PREVENT THE LIQUID HELIUM LEVEL FROM DROPPING TO A POINT, DURING RAMPING, WHERE A QUENCH MAY OCCUR.**

7. **MAKE SURE MAGNET VENTING IS INSTALLED IN CONFORMANCE WITH SET UP AND CALIBRATION, SECTION 1–2.**

8. **MAKE SURE THE COLD HEAD IS OPERATIONAL AND MAGNET SHIELD TEMPERATURES HAVE STABILIZED PRIOR TO RAMPING.**



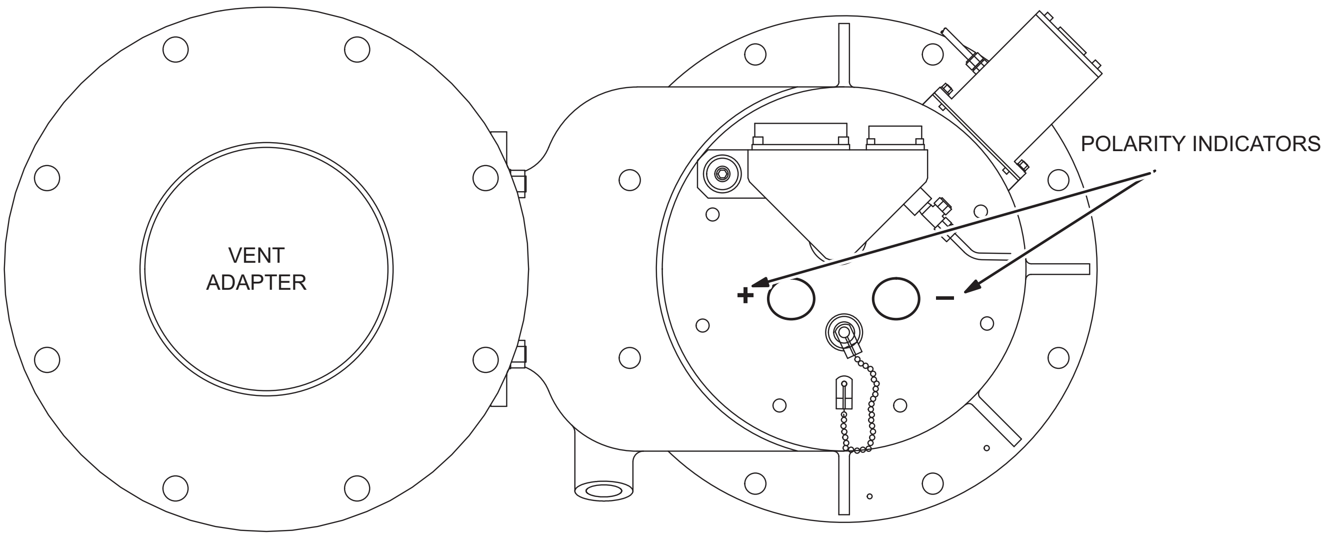
RAMP FLOWCHART
ILLUSTRATION 9–1

9–1 PREPARATION

CAUTION

The “Normal” ramp polarity is “+” on the left and “–” on the right as viewed from the cold head side of the magnet. “Reversed” ramp polarity has “+” on the right and “–” on the left as viewed from the cold head side of the magnet. If using a Polarity Plate, make sure that the Polarity Plate properly indicates which polarity the magnet is to be ramped. Make sure the chosen ramp polarity is recorded in the DATA SHEET tab, Section 6–1.

1. The normal Ramp Polarity is “+” on the left and “–” on the right, as viewed from the Cold Head side of the magnet. See Illustration 9–2.



POLARITY PLATE NORMAL CONFIGURATION
ILLUSTRATION 9–2

2. Set up the field monitoring equipment, Teslameter and Teslameter Probe, in conformance with SET UP AND CALIBRATION, Section 5.

9–1 PREPARATION (continued)

3. Perform Magnet Electrical Checks in conformance with FUNCTIONAL CHECKS, Section 3.
4. Make sure the magnet is at least 90% full of helium before Ramping the magnet.

WARNING!

THE SIV MAGNET CAN BE QUENCHED IF THE MAGNET POWER SUPPLY EXPERIENCES LARGE OUTPUT VOLTAGE FLUCTUATIONS AND/OR EXCESSIVE RIPPLE. MAKE SURE THE POWER SUPPLY IS ROUTINELY CALIBRATED AT AN APPROVED FACILITY.

5. Make sure that the Main Power Supply is installed, checked and adjusted in conformance with the Vendor Manual supplied with the unit.
6. Make sure the Shim Lead Assembly is "Engaged" in conformance with SET UP AND CALIBRATION, Section 6.
7. Make sure that Input Power Cable to the Power Supply is disconnected.
8. Connect the Power Supply to the magnet by making all cable connections in conformance with SET UP AND CALIBRATION, Section 8. ("Electrical Connections For Ramping And Shimming")
9. Record the Main Coil connection polarity in DATA SHEETS, Table 6–1.
10. Set all power supply heater switches to the OFF position. Set CURRENT ADJUST and VOLTAGE controls to 0 (full CCW).
11. Connect the Input Power Cable to the Main Power Supply.
12. Make sure He Vent Valve (V2) is closed.

9–2 RESISTANCE CHECKS

1. Check Switch Heater and Shim Coil resistances in conformance with FUNCTIONAL CHECKS, Section 6.
2. Make sure CURRENT ADJUST and VOLTAGE controls on the Main Power Supply are off (full CCW).

Note

All "Main Coil Driving Voltages" provided in this procedure will be equal in magnitude but opposite in polarity for "Reverse Ramped" magnets.

WARNING!

MAKE SURE MAIN HEATER SWITCH IS OFF DURING THE RESISTANCE CHECKS.

3. Set the MAIN POWER and POWER ON switches to ON (Switches located on both the front and back of the Main Power Supply).

9–2 RESISTANCE CHECKS (continued)

4. Set HEATER 2 SHIM AXIAL switch to 1 (on) and observe current rise in ammeter (800–820 mA) to verify circuit continuity. Make sure Main Heater Switch is off.
5. Connect a Digital Voltmeter (DVM) to the end of the Voltage Sense Leads.
6. Set CURRENT ADJUST COARSE control on power supply to maximum (full CW).
7. Observe the Main Power Supply Ammeter and slowly turn the VOLTAGE control (CW) to set 500A current through the Main Power Leads, Lead Extensions and persistent Main Switch.
8. Record the voltage reading on the (DVM) in the DATA SHEET tab, Table 6–1.

WARNING!

A VOLTAGE READING GREATER THAN 150 MILLIVOLTS AT 500 AMPS INDICATES UNACCEPTABLE INTERNAL CONTACT RESISTANCE OF THE LEAD EXTENSIONS. HIGHER RESISTANCES WILL ADD MORE HEAT TO THE MAGNET INCREASING BOILOFF AND POSSIBLY CAUSING A QUENCH DURING RAMPING.

9. Perform one or more of the bulleted steps below, as necessary, if the DVM voltage is greater than 150mV.

   - Wait approximately 1 minute with the current running, readings may drop as the Power Lead Extensions cool.
   - Tighten the nuts on top of the Hold Down Tool.
   - If the reading still exceeds 150 mV: turn the VOLTAGE and CURRENT ADJUST controls to zero (full CCW), turn off Magnet Power Supply input power, then check/tighten the bolts securing the Ramp Cables to the Power Supply and Ramp Leads Extensions. Lift and reseat the Ramp Leads. Repeat Steps 6 through 8.

10. Set the power supply VOLTMETER SELECT SWITCH to MAIN POWER SUPPLY position (This will display the output of the power supply monitored at the output lugs).

Note

A voltage less than 2.2V at 735A indicates acceptable system resistance. If the voltage exceeds 2.2V during the test, follow the procedures in Step 9 for adjusting contact resistance.

11. Gradually increase the CURRENT ADJUST controls to pass 735A through the Main Power leads, Lead Extensions and persistent Main Switch while observing the Power Supply Voltmeter. If the voltage exceeds 2.2V, then check/tighten the bolts securing the Ramp Cables to the Power Supply and Ramp Lead Extensions.
12. Turn the CURRENT ADJUST and VOLTAGE controls off (full CCW) and continue with the ramping procedure after completion of Step 11.

9–3 MODIFIED RAMP PROCEDURE (STEP RAMP)

DESCRIPTION

This procedure uses a “Step Ramp” method of ramping. The process is to park the magnet at 1.3T, allow a minimum of 4 hours to settle, then ramp the magnet up to 1.5T. This method will help minimize the potential for a magnet quench.

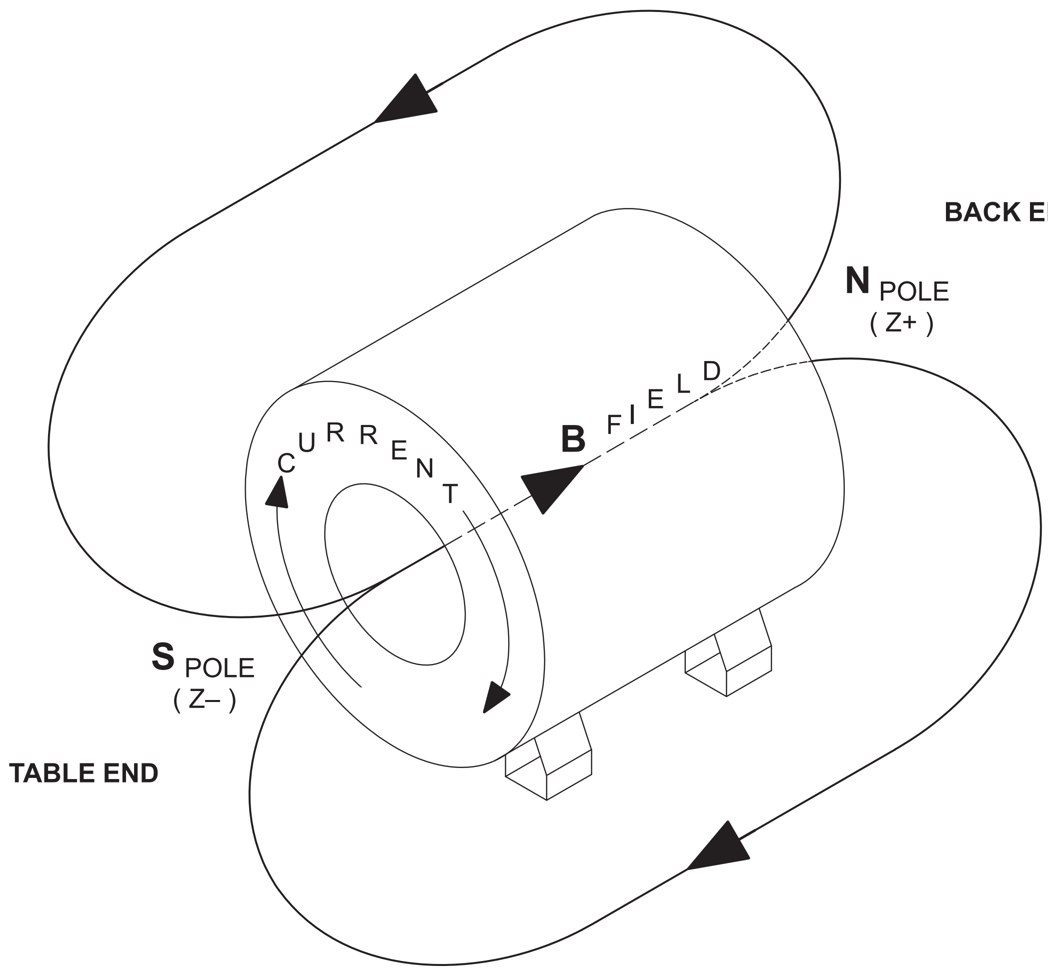
This ramp procedure is intended as a quench minimization tool and is a temporary measure that will only be used until a solution, for the quench issue, is in place. The field will be notified, when a solution is in place, through a Service Note.

All sections referred to in the following table are in Revision 0 of *Direction 15345,* GE 1.5T SIV Magnet & Cryogens Subsystem.

PRE RAMPING CHECKLIST

FUNCTION	NAME	DATE
HELIUM LEVEL MINIMUM 90%		
SHIM LEAD ENGAGED – SET UP AND CALIBRATION, SECTION 6		
FIELD MONITORING EQUIPMENT – SET UP AND CALIBRATION, SECTION 5		
RAMP SUPPLY CHECKED – SET UP AND CALIBRATION, SECTION 7		
RAMP LEAD CONTACT BANDS REPLACED – REPLACEMENT MAINTENANCE, SECTION 13		
ELECTRICAL CONNECTIONS FOR RAMPING AND SHIMMING – SET UP AND CALIBRATION, SECTION 8		
INSTALL RAMP LEAD HOLD DOWN FIXTURE – SET UP AND CALIBRATION, SECTION 7–2		
RESISTANCE CHECKS – SET UP AND CALIBRATION, SECTION 9–3		
RECORD MAGNET RAMP POLARITY – DATA SHEET TAB, TABLE 6–1		
HE VENT VALVE V2 CLOSED AND HELIUM GAS FLOWING OUT HOLES IN RAMP LEAD EXTENSIONS.		

9–3 MODIFIED RAMP PROCEDURE (STEP RAMP) (continued)



SIV MAGNET POLARITY FOR NORMAL RAMPED MAGNET
ILLUSTRATION 9–3

9–3 MODIFIED RAMP PROCEDURE (STEP RAMP) (continued)

PROCEDURE

CAUTION

If a Quench occurs during ramping, immediately turn VOLTAGE control and CURRENT control of the Main Coil Power Supply to zero (fully CCW). A quench is a rapid discharge of the magnetic field which will result in the rapid generation and expulsion of helium gas, rupturing the Burst Disc in the Vent System.

WARNING!

MAKE SURE THAT THE AXIAL SHIM SWITCH HEATERS ARE ON DURING RAMPING TO PREVENT COIL DAMAGE AND MAGNET QUENCH. GE POWER SUPPLIES HAVE PROTECTIVE CIRCUITRY TO PREVENT RAMPING VOLTAGE WITH THE AXIAL HEATER SWITCH IN THE “OFF” POSITION.

Note

Ice will form around the Ramp Lead Hold Down Tool Flow Holes during ramping. Remove ice as needed to maintain helium gas flow.

Note

The checklist is to aid in keeping track of progress, and should not be substituted for this procedure.

1. Set power supply VOLTAGE control to zero (full CCW).
2. Set power supply CURRENT ADJUST COARSE control to maximum (full CW).
3. Set HEATER 1 MAIN switch to 1 (on). Wait 3 minutes.
4. Set the power supply VOLTMETER SELECT SWITCH to MAIN COIL position.

Note

The main coil driving voltage will decay from the initial setting of 4.5 V. This is normal and no attempt to readjust the power supply voltage should be made until directed.

5. Adjust VOLTAGE control on Main Power Supply to display 4.5 Volts as indicated on the Power Supply Voltmeter (MAIN COIL position).
6. Estimate the system inductance by measuring current change over a 10 second ramping interval:

L(inductance) = 10 x Voltage/Current Change

Note

This method will give inaccurate values of inductance when the current is less than 200 Amps.

9–3 MODIFIED RAMP PROCEDURE (STEP RAMP) (continued)

Note

Measured inductance should be approximately 22.0 henrys(H), if the calculated value is between 20 – 24H continue with the procedure. If the calculated value is outside this range, discontinue ramping and measure Main Coil Resistance (See FUNCTIONAL CHECKS, Section 3). Contact the Region MAC Team Representative.

Note

The Teslameter will lock on when magnet current rises to about 350 amps. This usually occurs between 0.6 and 0.7 Tesla (25.545900 MHz – 29.803550 MHz).

7. When the magnet field reaches 1.0 T (42.576 MHz), slowly adjust VOLTAGE control on Main Power Supply to 2.0 Volts, as indicated on the power supply voltmeter. Make sure the Power Supply Voltmeter is set in the MAIN COIL position. The magnet current will be around 510 amps at this time.

WARNING!

THE SIV MAGNET CAN BE QUENCHED IF LARGE CURRENT OR VOLTAGE CHANGES OCCUR NEAR PARKING. WHEN ADJUSTING THE VOLTAGE IN STEPS 8 AND 10 BELOW, TURN THE VOLTAGE CONTROL IN A SLOW, CONTINUOUS MOTION. BE CAREFUL NOT TO JERK THE VOLTAGE CONTROL.

Note

The magnet is to be parked at 1.3T (55.35 MHz), allowed to settle for 4 hours, then ramped up to final parking field.

8. As the magnet field approaches 1.3T(55.35 MHz), slowly decrease the main coil driving voltage (VOLTAGE ADJUST control) to obtain a reading of 0.00 volts across the main coil as indicated on the power supply voltmeter (MAIN COIL position).

9. Maintain field at final setting for 5 minutes before proceeding to Step 10.

Note

Observe voltage reading on Power Supply Voltmeter with toggle switch in MAIN COIL position. When the switch goes “persistent” the voltage across the magnet terminals will stabilize at 0.00.

Note

Magnet Field may decay slightly after main switch heater is turned off. Maintain Teslameter reading constant by slowly adjusting VOLTAGE ADJUST control.

9–3 MODIFIED RAMP PROCEDURE (STEP RAMP) (continued)

10. Turn off Main Switch Heater. Wait 8 to 10 minutes for the switch to fully cool and go “persistent”.
11. Record the Current value at which the switch went “persistent” _______________________.

Note

Check that Teslameter does not decrease as the VOLTAGE control knob is turned to minimum (Full Counterclockwise). Only the last two digits on the Teslameter should change. If the field decreases as the VOLTAGE control knob is turned, the main coil switch is not persistent and the VOLTAGE control must be slowly adjusted to return to Parking Field. The field will drop approximately 1 KHz when the power supply is being dialed down to zero amps.

12. When the switch goes “persistent”, slowly turn the power supply VOLTAGE ADJUST control to minimum (Full Counterclockwise) over a two minute period.
13. Gradually turn the CURRENT ADJUST controls to zero (over a one minute period).
14. Set HEATER 2 SHIM AXIAL switch to 0 (off).
15. Set MAIN POWER AND POWER ON switches to OFF and disconnect Input Power Cable.
16. Open Vent Valve (V2) to de–pressurize the Cryostat to 0.25 psig. Close V2.
17. Remove Ramp Lead Hold Down Tool from magnet.
18. Replace Flow Hole Screws in Ramp Lead Extensions.
19. Remove all ice around Ramp Lead Port Compression Nuts.
20. Unscrew the Ramp Lead Port Compression Nuts and remove Ramp Lead Extensions. Immediately replace the caps onto the Ramp Lead Ports.
21. Let magnet stabilize for a minimum of 4 hours before continuing with Step 22.
22. Perform the steps in the following table:

FUNCTION	NAME	DATE
ELECTRICAL CONNECTIONS FOR RAMPING AND SHIMMING – SET UP AND CALIBRATION, SECTION 8		
INSTALL RAMP LEAD HOLD DOWN FIXTURE – SET UP AND CALIBRATION, SECTION 8–2		
RESISTANCE CHECKS – SET UP AND CALIBRATION, SECTION 9–3		
HE VENT VALVE V2 CLOSED AND HELIUM GAS FLOWING OUT HOLES IN RAMP LEAD EXTENSIONS.		

23. Make sure the Axial Shim Heater is on and Ramp Leads are connected with the polarity indicated in Table 6–1 of the Data Sheet tab.
24. Set the power supply voltmeter toggle switch to MAIN COIL position.
25. Set the power supply CURRENT ADJUST controls to maximum (Full Clockwise).

9–3 MODIFIED RAMP PROCEDURE (STEP RAMP) (continued)

26. Set VOLTAGE ADJUST control to adjust power supply output current to the value obtained in Step 11 above.

27. Turn on the Main Switch Heater. Leave the Axial Shim Switch Heater Supply on until directed otherwise.

28. Allow approximately 3 minutes for the Main Switch to go normal.

Note

Drive voltage will decay as magnet field increases. Do not attempt to adjust drive voltage until directed.

29. When the switch goes normal, slowly adjust the VOLTAGE ADJUST control for a main coil drive voltage 1.0 volts.

30. As the magnet field approaches 1.48 T (63.013 MHz), slowly adjust VOLTAGE control on main power supply to 150 millivolts (0.150 volts) as indicated on the power supply voltmeter (MAIN COIL position).

Note

The specification for the Magnetic Field is15000.0 to 15002.0 gauss (63.866 – 63.874 MHz).

Note

The magnet parking field will normally increase during Mechanical Shim and Quickshim. The average amount of field change is between 14 gauss (59.607 KHz) and 15 gauss (63.865 KHz) and is dependent on the amount and location of steel in the room. Parking the magnet lower, by this average amount (between 14 gauss and 15 gauss), may eliminate the need for field adjustment after Mechanical and “Quickshim” shimming.

Note

For Sierra System Magnets, adjust the Magnet Field, in Steps 10 and 11, to 14962 – 14964 gauss (63.64 – 63.76 MHz).

31. As the magnet field approaches the parking field, between 14,986 to 14,987 gauss (63.805 – 63.809 MHz), slowly decrease the main coil driving voltage (VOLTAGE control) to zero as indicated on the power supply voltmeter (MAIN COIL position).

32. Check Teslameter and slowly adjust the VOLTAGE control, as required, to bring Magnetic Field between 14,986 to 14,987 gauss (63.805 – 63.809 MHz). The total current will be approximately 740 amps. Allow final field to stabilize. The last two digits on the Teslameter should be the only digits changing.

33. Maintain field at final setting for 5 minutes before proceeding to Step 34.

9–3 MODIFIED RAMP PROCEDURE (STEP RAMP) (continued)

Note

Observe voltage (read on Power Supply Voltmeter with toggle switch in MAIN COIL position). When the switch goes "persistent" the voltage across the magnet terminals will stabilize at 0.00.

Note

Magnet Field may decay slightly after main switch heater is turned off. Maintain Teslameter reading constant by slowly adjusting VOLTAGE ADJUST control.

34. Turn off Main Switch Heater. Wait 8 to 10 minutes for the switch to fully cool and go "persistent".

35. Record current value at which the switch went "persistent" in DATA SHEETS, Table 6–1.

WARNING!

MAKE SURE THAT THE CONNECTION POLARITY AND FINAL RAMPING CURRENT ARE RECORDED IN DATA SHEETS, TABLE 6–1. THIS INFORMATION IS ESSENTIAL FOR LATER CHANGING OF THE MAGNETIC FIELD. THE MAIN POWER SUPPLY MUST BE SET TO THE SAME CURRENT AND POLARITY IN THE MAIN COILS TO AVOID A QUENCH WHEN TURNING ON THE MAIN SWITCH.

Note

Check that Teslameter does not decrease as the VOLTAGE control knob is turned to Zero. Only the last two digits on the Teslameter should change. If the field decreases as the VOLTAGE control knob is turned, the main coil switch is not persistent and the VOLTAGE control must be slowly adjusted to return to Parking Field. The field will drop approximately 1 KHz when the power supply is being dialed down to zero amps.

36. When the switch goes "persistent", slowly turn the power supply VOLTAGE control to zero over a two minute period (Full CCW).

37. Gradually turn the CURRENT ADJUST control to zero (over a one minute period).

38. Set HEATER 2 SHIM AXIAL switch to 0 (off).

39. Set MAIN POWER AND POWER ON switches to OFF and disconnect Input Power Cable.

9–3 MODIFIED RAMP PROCEDURE (STEP RAMP) (continued)

40. Open Vent Valve (V2) to de–pressurize the Cryostat to 0.25 psig. Close V2.
41. Remove Ramp Lead Hold Down Tool from magnet.
42. Replace Flow Hole Screws in Ramp Lead Extensions.
43. Remove all ice around Ramp Lead Port Compression Nuts.
44. Unscrew the Ramp Lead Port Compression Nuts and remove Ramp Lead Extensions. Immediately replace the caps onto the Ramp Lead Ports.

Note
Transverse coil currents will be removed in SET UP AND CALIBRATION, Section 10–2.

45. There may be small currents induced onto the Transverse Coils by the Ramping process. Make sure these currents are removed in conformance with SET UP AND CALIBRATION, Section 10–2 before Magnetic Probe Centering and Shimming the magnet.

CAUTION

Cryostat exhaust flow rates and pressure must be checked and adjusted as required after magnet installation, ramping and shimming to ensure that proper cooling conditions are maintained and no leaks are present in the Helium Exhaust System or Vent Valve (V2).

Note
Read all flow rates from the bottom of the float (ball) on the flow meters.

Note
Flow rates may be temporarily elevated after ramping. Do not adjust them until after the magnet has had time to stabilize (at least one day).

46. Make sure the following conditions are maintained. Re–check settings in three days and again after one week:

INSTRUMENTATION LEAD FLOWMETER (F1) = 0.8 – 1.2 SCFH
SHIM LEAD FLOWMETER (F2) = 1.8 – 2.2 SCFH
CRYOSTAT GAUGE PRESSURE = 0.25 – 0.50 psig

Note
If flow rate through F2 is less than 1.8 SCFH or the pressure gauge reads less than 0.25 psig, pressurize the vessel and “bubble test” all exhaust plumbing joints, relief valve and Shim Lead Connector. Make sure V2 is fully closed. Repair any leaks.

47. Proceed to SET UP AND CALIBRATION, Section 10 (“Shimming Preparation/Field Stabilization”).

9–3–1 Step Ramp Checklist

STEP RAMP CHECKLIST

Please return this completed checklist to :

GE Medical Systems
P.O. Box 100539
Florence South Carolina
29501–0539
Attn: Bob Behl
Mail Code 105

MAGNET SERIAL NUMBER____________________

1. Switch on Axial and Main Heaters. Wait three minutes. **Time:**_______________.
2. Adjust VOLTAGE ADJUST control for 4.5 volts magnet drive voltage and allow to decay.
3. When 1.0T field is obtained, adjust main coil driving voltage to 2.0 volts and allow to decay.
4. When 1.3T field is obtained, park magnet and record parking current polarity and value.
   Time:_______________.
5. Remove ramp lead extensions and allow magnet to stabilize for 4 hours.
6. Re connect power supply, ramp lead extensions and hold down tool.
7. Perform ramp lead resistance checks. **Ramp Lead Voltage Drop__________________mV.**
8. Switch on Axial Heater. **Time:**_______________Wait three minutes.
9. Set power supply to Park Current value in Step 4 above.
10. Switch on Main Heater. **Time:**_______________Wait three minutes.
11. After switch goes normal, adjust main coil drive voltage to 1.0 volts.
12. When 1.48T field is obtained, adjust main coil drive voltage to 150 millivolts (0.150V).
13. Park magnet at desired field and record Parking Current and Parking Field in Table 6–1 of the Data Sheet tab.

    Time:_________________________.

    Final Field:______________________________.

    Parking Current:_________________________.
14. Reduce vessel pressure to less than 1.0psig and remove the power leads.